

FOODGRID AI — PHASE 2

Existing Solutions, Organizations, and the Emerging Research Gap

Purpose: Identify what already exists before claiming FoodGrid AI is novel. This phase is a literature and ecosystem review, not a claim that FoodGrid has partnerships.

1. Why Existing-Solution Research Matters

Food redistribution is already an established field. FAO describes recovery and redistribution as an ecosystem involving food banks, social supermarkets, food pantries, soup kitchens, shelters, community programmes, and other initiatives. Therefore, FoodGrid AI should not be presented as the first system to connect surplus food with people or organizations. The research question is instead what additional coordination, optimization, data integration, or global interoperability a new system could contribute.

2. Major Existing Models

Existing model	What it does	Evidence / limitation relevant to FoodGrid
Too Good To Go	Consumer marketplace plus business software.	Shows that surplus food can be sold through a marketplace. It also demonstrates that surplus food can be sold through a marketplace. It also demonstrates that surplus food can be sold through a marketplace.
OLIO	Community redistribution using local volunteers.	Shows that surplus food can be redistributed through a community-based model. It also demonstrates that surplus food can be redistributed through a community-based model.
Feeding America / MealConnect	US charitable food network with a digital food-coordination platform.	Shows that a large-scale food redistribution network already exists. It also demonstrates that a large-scale food redistribution network already exists.
FoodCloud / Foodiverse	Connects food businesses with charities through a digital platform.	Shows that a digital platform for food redistribution already exists. It also demonstrates that a digital platform for food redistribution already exists.

3. Detailed Findings

Too Good To Go. The company describes its app as a large surplus-food marketplace and also offers business software. Its platform includes digital near-expiry management, AI-powered discounting, consumer sales, and charity redistribution. This means FoodGrid should not claim that simply adding AI to surplus-food management is novel. A stronger contribution would be a research-tested coordination model combining donation, demand, transport, time constraints, and multi-stakeholder optimization.

OLIO. OLIO's Food Waste Heroes programme uses trained local volunteers to collect and redistribute surplus from businesses. The programme is explicitly designed to fill gaps where charity collections may not cover every retailer. This suggests that FoodGrid should model both professional logistics and volunteer capacity as possible transport resources, while maintaining safety and traceability requirements.

Feeding America / MealConnect. Feeding America reports that MealConnect has facilitated billions of pounds of food rescue and is integrated into a large U.S. charitable-food network. This demonstrates the importance of network effects: a platform becomes useful when donors and recipient organizations are already connected. FoodGrid therefore needs an onboarding, verification, and capacity model rather than only a matching algorithm.

FoodCloud / Foodiverse. FoodCloud is especially important for the research because its model is close to FoodGrid's proposed architecture. FoodCloud reports direct retailer-to-charity matching, regional hubs, refrigerated collection, warehouse management, and international technology deployment. Its experience shows that real-world redistribution depends on food safety, traceability, local demand, transport capacity, and hub economics. FoodGrid should use FoodCloud as a major benchmark in the literature review.

4. Research Literature on Food Rescue

A systematic review by Hecht and Neff examined evaluations of food-rescue interventions and noted the importance of assessing and comparing different rescue models rather than assuming that all interventions produce the same outcomes. Earlier optimization research by Phillips, Hoenigman and Higbee modeled food rescue and redistribution using simulation and highlighted the importance of expiration speed, transport, supplier density, and operational resources. More recent research has also investigated automated allocation and fairness in food-drop matching. Together, these studies support making FoodGrid a measurable optimization and simulation project rather than only a website prototype.

5. What FoodGrid AI Must NOT Claim

- It should not claim to be the world's first food-rescue platform.
- It should not claim that AI alone solves hunger or food waste.
- It should not list major companies as FoodGrid partners unless an actual partnership exists.
- It should not claim real-world impact metrics before field data exists.
- It should not treat computer vision as a substitute for food-safety inspection.

6. Candidate Research Gap

The initial review suggests that the broad donor-to-recipient digital connection is already well established. The research gap should therefore be narrower and more defensible. A promising direction is to study a multi-layer coordination model in which surplus supply, recipient demand, time-to-expiry, transport capacity, route cost, and fairness are optimized together. The research can then test whether this integrated model performs better than simpler baselines.

Proposed gap statement:

“Existing food-rescue systems demonstrate the feasibility of digital surplus-food redistribution, but there remains a research opportunity to evaluate an integrated, multi-stakeholder optimization framework that jointly models supply, demand, perishability, transport capacity, geographic constraints, and measurable impact across multiple local networks.”

7. FoodGrid AI Research Differentiation

Layer	Existing examples	FoodGrid research direction
Supply capture	Business uploads / lists surplus	Unified donor schema + structured metadata
Demand	Connected charities / local recipients	Explicit demand forecasting and capacity representation
Matching	Notifications / connected networks	Constraint-based multi-factor matching
Perishability	Date/near-expiry tools and safety rules	Research model for remaining usable window with conservative rules
Transport	Volunteers, hubs, direct pickup, fleets	Joint assignment + route optimization
Fairness	Network-specific policies	Test whether allocation can balance efficiency and recipient fairness
Impact	Platform-specific reports	Reproducible methodology for food, logistics, cost, and environmental m
Global interoperability	Some international expansion exists	Study a common data/API architecture across local networks

8. Phase 2 Conclusion

The review confirms that FoodGrid is entering an active and already-developed field. This is positive for the project because it provides real systems to learn from and real research questions to test. The strongest path forward is not to compete on a generic claim of 'connecting donors and NGOs,' but to build and evaluate a technically explicit coordination layer that integrates demand, perishability, matching, transport, fairness, and impact measurement.

9. Initial Source List

- [1] Too Good To Go. Save Good Food From Going To Waste. <https://www.toogoodtogo.com/en-us/>
- [2] Too Good To Go. Too Good To Go Platform. <https://www.toogoodtogo.com/en-gb/platform>
- [3] OLIO. Food Waste Heroes Programme. <https://olioapp.com/business/food-waste-heroes-programme/>
- [4] Feeding America. Food Rescue Fund / MealConnect. <https://www.feedingamerica.org/partners/food-rescue-fund>
- [5] Feeding America. MealConnect: 5 Billion Pounds Rescued. <https://www.feedingamerica.org/about-us/press-room/mealconnect-5-billion-pounds-rescued>
- [6] FoodCloud. FAQs / Foodiverse / Redistribution. <https://www.food.cloud/faqs>
- [7] FoodCloud. Get Good Food. <https://www.food.cloud/get-food>
- [8] FoodCloud. About Us / History. <https://www.food.cloud/about>
- [9] Hecht, A. A., & Neff, R. A. (2019). Food Rescue Intervention Evaluations: A Systematic Review. Sustainability, 11(23), 6718. <https://doi.org/10.3390/su11236718>
- [10] Phillips, C., Hoenigman, R., & Higbee, B. (2011). Food Redistribution as Optimization. arXiv:1108.5768. <https://arxiv.org/abs/1108.5768>
- [11] Mertzaniadis, M., Psomas, A., & Verma, P. (2024). Automating Food Drop: The Power of Two Choices for Dynamic and Fair Food Allocation. arXiv:2406.06363. <https://arxiv.org/abs/2406.06363>

Next phase: Phase 3 will turn these findings into a formal research gap, novelty statement, hypothesis, and measurable FoodGrid contribution.